Year 12 Further Maths - Matrices

Similar questions - Algebra, inverses and simultaneous equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & -1 & 2 \\ 2 & 1 & a \end{pmatrix}$.

(a) Show that $\det \mathbf{A} = 11 - 7a$. **[3]**

(b) State the value of a for which $\mathbf{A}$ is singular. **[1]**

(c) Given that $\mathbf{A}$ is non-singular, find $\mathbf{A}^{-1}$ in terms of a. **[5]**

2 The matrices $\mathbf{A}$ and $\mathbf{B}$ are given by $\mathbf{A} = \begin{pmatrix} t & 4 \\ 2 & t \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} t & 2 \\ 4 & t \end{pmatrix}$, where t is a constant.

(a) Show that $|\mathbf{A}| = |\mathbf{B}|$. **[2]**

(b) Verify that $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$. **[3]**

(c) Given that $|\mathbf{AB}| = 1$, find all possible values of t. **[3]**

3 Use a matrix method to determine the solution of the simultaneous equations

$$2x + y - z = 0, \qquad x + 3y + 2z = 5, \qquad 3x - y + z = 10.$$

[5]

4 The matrices $\mathbf{A}$ and $\mathbf{B}$ are given by $\mathbf{A} = \begin{pmatrix} a & 2 \\ -1 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 4 & -1 \\ 0 & 2 \end{pmatrix}$, where a is a constant.

(a) Find $\mathbf{A} + \mathbf{B}$, $\mathbf{AB}$ and $\mathbf{A}^2$. **[3]**

(b) Given that $\det \mathbf{A} = 25$, find a. **[2]**

(c) You are given instead that the system $ax + 2y = 1$, $-x + 3y = 4$ does not have a unique solution. Determine the value of a and state whether the equations are consistent. **[3]**

Question 1

(a) $\det\mathbf{A} = 1(-a-2) - 2(3a-4) + 1(3+2) = -a-2-6a+8+5 = 11-7a$

(b) $a = \frac{11}{7}$

(c) Cofactors give the adjugate $\begin{pmatrix} -a-2 & 1-2a & 5 \\ 4-3a & a-2 & 1 \\ 5 & 3 & -7 \end{pmatrix}$, so $\mathbf{A}^{-1} = \dfrac{1}{11-7a}\begin{pmatrix} -a-2 & 1-2a & 5 \\ 4-3a & a-2 & 1 \\ 5 & 3 & -7 \end{pmatrix}$

Question 2

(a) $|\mathbf{A}| = t^2-8$, $|\mathbf{B}| = t^2-8$

(b) $\mathbf{AB} = \begin{pmatrix} t^2+16 & 6t \\ 6t & t^2+4 \end{pmatrix}$, $|\mathbf{AB}| = (t^2+16)(t^2+4) - 36t^2 = t^4-16t^2+64 = (t^2-8)^2 = |\mathbf{A}||\mathbf{B}|$

(c) $(t^2-8)^2 = 1 \Rightarrow t^2-8 = \pm 1 \Rightarrow t^2 = 9$ or 7, so $t = \pm 3$ or $\pm\sqrt{7}$

Question 3

$\mathbf{M} = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \\ 3 & -1 & 1 \end{pmatrix}$, $\det\mathbf{M} = 25$, $\mathbf{M}^{-1} = \dfrac{1}{25}\begin{pmatrix} 5 & 0 & 5 \\ 5 & 5 & -5 \\ -10 & 5 & 5 \end{pmatrix}$. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{M}^{-1}\begin{pmatrix} 0 \\ 5 \\ 10 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$: $x = 2$, $y = -1$, $z = 3$

Question 4

(a) $\mathbf{A}+\mathbf{B} = \begin{pmatrix} a+4 & 1 \\ -1 & 5 \end{pmatrix}$, $\mathbf{AB} = \begin{pmatrix} 4a & 4-a \\ -4 & 7 \end{pmatrix}$, $\mathbf{A}^2 = \begin{pmatrix} a^2-2 & 2a+6 \\ -a-3 & 7 \end{pmatrix}$

(b) $\det\mathbf{A} = 3a+2 = 25$, $a = \frac{23}{3}$

(c) No unique solution when $\det = 0$: $3a+2 = 0$, $a = -\frac{2}{3}$. Then the first equation is $-\frac{2}{3}x+2y = 1$, equivalent to $-x+3y = \frac{3}{2}$, which contradicts $-x+3y = 4$: the equations are inconsistent (no solutions)